



The impact of supply chain finance on firm cash holdings: Evidence from China

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ARTICLE INFO

Keywords:

Supply chain finance
Core firm
Cash holdings
Competitiveness
Investment efficiency

ABSTRACT

We establish a link between supply chain finance (SCF) and a firm's cash holdings through a proprietary dataset of listed firms on the Shanghai and Shenzhen Stock Exchanges. Our findings suggest that SCF may significantly improve cash holdings by a core firm, speed up its cash turnover, and enhance its competitiveness. Such impacts are stronger when the firm is of lower efficiency. In addition, a firm with more commercial credit or of state ownership is more likely to become committed to SCF. A close relationship with banks or political connections may further improve the private firm's cash holdings. In the meantime, increased cash holdings at the core firm may lead to enhanced investment efficiency, rather than over-investment, and enhanced firm values. Our findings may have such policy implications that SCF needs to be better supported by the banking sector and the government for China's sustainable economic success.

1. Introduction

Since the Global Financial Crisis (GFC) in 2008, supply chain finance (SCF) has emerged as an efficient means of short-term financing to alleviate a firm's financial constraint (Nienhuis et al., 2013; Ali et al., 2019; Jia et al., 2020). Often with better access to bank loans and enjoying more commercial credit, a core firm in SCF can share its financial resources with other firms in the supply chain or provide guarantees to facilitate small and medium-sized enterprises (SMEs) to borrow bank loans. Given the outstanding lending preference towards state-owned enterprises (SOEs) (Xu and Lin, 2007; Fu et al., 2015; Xu et al., 2015), and the traditional bias against either SMEs or those of private ownership (Allen et al., 2005; He et al., 2019; Li et al., 2020a; Li et al., 2020b; Wu and Xu, 2020), SCF has been growing rapidly in the largest bank-based economy (Lin and Zhang, 2020; Song et al., 2020).

Through SCF, the core firm may effectively support an ecosystem of suppliers, manufacturers, distributors, and customers with better controls of capital, logistics, and information capital flows (Danny et al., 2010). The optimised capital flow within the supply chain may be realised with support from banks (Camerinelli, 2009; Lamoureux and Evans, 2011) and cash flow management can be enhanced when capital and product flows are aligned (Wuttke et al., 2013a). Closely tied to upstream and downstream stakeholders, the core firm may efficiently allocate funds from banks and other creditors to various firms in the supply chain, especially the SMEs, and subsequently improve the competitiveness of the supply chain. To facilitate SCF development, the State Council, the People's

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<https://doi.org/10.1016/j.pacfin.2020.101402>

Received 11 June 2020; Received in revised form 20 July 2020; Accepted 27 July 2020

Available online 01 August 2020

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Bank of China (PBC), China Banking and Insurance Regulatory Commission (CBIRC), China Securities Regulatory Commission (CSRC), Ministry of Finance (MOF), and Ministry of Commerce (MOC), have all intensively issued relevant policies in recent years.¹

The literature on SCF has been evolving in two strains. The finance strain commonly defines SCF as a short-term financing solution (Camerinelli, 2009; Lamoureux and Evans, 2011; More and Basu, 2013). As an important financial innovation bridging bank loans and firms in need, SCF may serve the purpose of simultaneous management of a core firm's working capital and that of the supply chain with such a mechanism as reverse factoring (Lekkakos and Serrano, 2016). In addition, SCF may effectively improve capital efficiency during times of a credit crunch (Polak et al., 2012). Through reduced transaction cost and default risks, SCF may optimise efficiency of the supply chain, where SMEs play important roles (Ali et al., 2019). In comparison, the supply chain strain focuses on the optimisation of the ecosystem (Gomm, 2010; Wuttke et al., 2013b). SCF may contribute to framework design and risk management in particular sectors (Hofmann and Belin, 2011). Integrating the financial needs of suppliers, manufacturers, distributors, and customers, SCF provides an optimal solution (Hofmann, 2014). The integration of material, information, and capital flows, in subsequence, contribute to improved firm performance (Danny et al., 2010). However, there is a growing need for further empirical and theoretical discussions on the core firm with reference to its sector's features, such as its behaviour and performance (Gelsomino et al., 2016). Given the voluntary practice of information disclosure on supply chains in most regulatory regimes, there has been scarce empirical literature on core firms in SCF.

In this study, we examine the impact of SCF on a core firm's cash holdings through a proprietary dataset of listed firms on the Shanghai and Shenzhen Stock Exchanges. Our findings suggest that SCF may significantly improve cash holdings by the core firm, speed up its cash turnover, and enhance its competitiveness. These impacts are stronger when the firm is of lower efficiency. In addition, a firm with more commercial credit or of state ownership is more likely to become committed to SCF. A closer relationship with banks or political connections may further improve the private firm's cash holdings. In the meantime, increased cash holdings at a core firm may lead to enhanced investment efficiency, rather than over-investment, and enhanced firm values. In other words, SCF may play positive roles in improving the firm's efficiency and competitiveness and be meaningful for the development of real economy.

Our study may be meaningful in at least four aspects. First, this empirical study may contribute to the evolving literature on SCF. Existing studies have largely ignored the SCF impact on a core firm's financial behaviour. In the meantime, literature on cash holdings often focuses on ordinary firms without a SCF focus. As a result, the link between SCF and a core firm's cash holdings may be a helpful extension to the literature. Second, the mechanism of SCF may be further explored in the context of the largest emerging market. Given the noticeable growth of SCF in recent years, such exploration may be useful to better understand this impactful financial innovation, especially for the supply chains connecting large numbers of SMEs with real economy. The recent Trade War in 2018 and the Covid-19 Pandemic in 2020 may have imposed tangible financial pressure on supply chains in the world economy. Such an examination may assist an understanding of a core firm's financial status and subsequent functioning in the restoration of supply chains. Third, the listed firms on Shanghai and Shenzhen Stock Exchanges often take a lead in reforming the SOEs in the largest bank-based economy. Given the preference of banks for lending to SOEs and the outstanding financial constraints of private firms or SMEs, such a discussion of SCF impact on firms of various ownership and size may shed light on effective financial solutions in future studies. Fourth, our findings may have policy implications in the emerging market contexts. SCF may have an impact on a firm's cash holdings, so the industrial and banking lending policies may need adjustments to further support SCF and the real economy.

The rest of the paper is structured as follows. Part 2 reviews the literature and develops our research hypotheses. Part 3 outlines data and methodology. Part 4 analyses our results and Part 5 concludes the paper.

2. Literature review and hypotheses development

The core firm often plays key roles in the supply chain. Besides designing the SCF, the core firm also provides credit and information to other firms. The SCF-based network allows better management of material, information, and cash flows (Liebl et al., 2016; Wuttke et al., 2019). The core firm may organically integrate finance and business operations among firms in the supply chain (Lin and Zhang, 2020; Song et al., 2020). The competitiveness of firms, including SMEs, can be significantly enhanced through optimal inter-firm capital flows and efficiency of cash flow management (Wuttke et al., 2013a). The cash flow management in SCF may lead to a different level of cash holdings at the core firm. Evidence from developed markets suggests that SCF may shorten the period of accounts receivable while increasing that of accounts payable (Hofmann and Kotzab, 2010). As a result, SCF may enhance the efficiency of capital flows and demonstrate higher cash holdings at the core firm (Lamoureux and Evans, 2011).

Given the features of bank-based economic fundamentals (Levine, 2002; Allen et al., 2005), the emergence of SCF and subsequent cash holdings by the core firm may be largely related to the core firm's ties with its major creditors, especially the banks. On one hand, the core firm in SCF may demonstrate better risk control through such innovative financial solutions as factoring and reverse factoring, which reduce the risk of bankruptcy in the supply chain (Klapper, 2006; Lekkakos and Serrano, 2016). On the other, banks may infiltrate more borrowers through accurate information from the core firm in SCF (Caridi et al., 2010; Hofmann, 2014). Under noticeable information asymmetry, banks in China are often reluctant to lend to private firms or SMEs (Wu and Xu, 2020; Li et al.,

¹ For example, PBC (2016) No. 42 Document, "Several opinions of the People's Bank of China, the National Development and Reform Commission, the Ministry of Industry and Information Technology, and other authorities on financial support for the industrial sector in stabilizing growth, adjusting structure and heightening benefits", was issued on 14 February 2016. Another example can be State Council (2017) No. 84 Document on "Active guidance on supply chain innovation and application" issued on 5 October 2017.

2020a). Innovative financial solutions within the core firm may also contribute to increased commission fees for the banks besides better risk management of loans to the supply chain. Such a new relationship with firms in the supply chain may be preferred by the fast expanding Chinese banks since the GFC, as they previously competed against each other with similar products and services (Allen et al., 2005; Xu and Lin, 2007; Xu et al., 2015). In addition, since interest income is the most important source of income for Chinese banks, there may be incentives for them to push their lending to the core firm in SCF. As a result, the core firm may increase its cash holdings to ensure sufficient liquidity in the supply chain.

Based on the above theoretical discussion, we develop our first hypothesis as follows.

Hypothesis 1. *Ceteris paribus*, SCF positively contributes to more cash holdings at the core firm.

However, excessive cash holdings may also lead to various agency issues. Firm executives may take advantage of excessive cash holdings for personal benefits and subsequently destroy firm values (Myers and Rajan, 1998; Pinkowitz et al., 2006; Dittmar and Marhrt-Smith, 2007). On the contrary, insufficient cash holdings may induce liquidity issues in the supply chain with increased risk of bankruptcy. As a result, the core firm may need to achieve a dynamic balance between the marginal earnings and cost of cash holdings. In other words, the optimal cash holdings by the core firm in SCF may be affected by its economic consequences. When cash holdings contribute more marginal revenue than marginal cost, the core firm tends to increase its cash holdings. As a result, the core firm in SCF may need to consider its optimal cash holdings to meet its various targets.

Given the slowing market and increased competition among Chinese firms in recent years (He et al., 2019; Wu and Xu, 2020), SCF may need to better enhance the competitiveness of firms within the supply chain (Wuttke et al., 2013b; Wuttke et al., 2019). Existing studies often suggest that cash holdings often inevitably impact on a firm's competitiveness in the marketplace. Sufficient cash holdings may enable the firm to better execute its strategies, such as R&D investments, mergers and acquisitions (M&As), or predatory pricing, and take more market share away from its competitors (Haushalter et al., 2007; Fresard, 2010). Similarly, the core firm in SCF may take advantage of its financial network, including its ties to banks, to gain comparative advantages exclusive of its competitors. SCF may improve trust, commitment, and profitability of firms in the supply chain, which subsequently improves their access to bank loans and lowers funding costs (Randall and Farris, 2009; Camerinelli, 2009; Lamoureux and Evans, 2011). The lowered funding costs among firms in the supply chain may further improve the competitiveness of the core firm.

Based on the above theoretical discussion, we develop our second hypothesis as follows.

Hypothesis 2. *Ceteris paribus*, improved cash holdings positively contribute to improved competitiveness of the core firm.

3. Sample and research methodology

3.1. Sample

We adopt all listed firms on the Shanghai and Shenzhen Stock Exchanges between 2011 and 2017 as our initial sample. We first manually collect SCF information from the firms' financial reports, public announcements, and Baidu searches. Excluding financial firms and those without SCF commitment, we have 344 sample firms. Then we further exclude firms from the network and information industry,² those starting SCF in 2017, those without a clear record of SCF starting year, or those without a clear business scope. Eventually, we have 118 effective sample core firms, which play key roles in industrial chains and are core firms providing financial support to their up- and down-stream firms in SCF. Table 1 reports the distribution of these sample firms. We further use firm size, internal control quality, governance, profitability, leverage, capital expenditure, non-cash assets, sales and year as controls for the closest propensity score matching (PSM) at a ratio of 1:4. We get 438 listed firms without SCF commitment in our comparison group.³ Including 118 core firms in SCF, we have 556 listed firms altogether in our whole sample (see Appendix).

We derive financial data of the 556 listed firms from CSMAR and WIND databases. Where there is data missing from the databases, we apply manually collected data from the firms' financial statements. We also collect firm internal control data from the DIB database.⁴ We then winsorize the collected data at 1% significance level to remove extreme values.

3.2. Research methodology

Cash holdings and competitiveness of firms would be the key variables in this study. Following Kalcheva and Lins (2007), we define cash holdings (*Zcash*) as (cash + short-term investment or trading securities)/total assets. To identify change in cash holdings, we adopt $\Delta Zcash$ as a dependent variable in this study. $\Delta Zcash$ = (cash holdings in the year after SCF commitment – cash holdings in

² Given the nature of their business scope, most firms in the network and information sector are service providers to the core firms in other sectors. As a result, they are not identified as core firms in SCF.

³ Our comparison group has ATT values lower than 1.69. *t*-test results also suggest that core firms in SCF and comparison samples are not significantly different in the selected control variables. The data balance test results also suggest that the samples for comparison purpose are acceptable. In addition, it should be noted that the 438 control sample enterprises do not include the excluded samples from the 344 original enterprises.

⁴ The DIB database provides an internal control index of listed Chinese firms. In conformity with the regulatory requirements, the DIB index is developed on the listed firm's internal control evaluation reports, internal control audit reports and annual reports. The DIB database has been accepted by a growing number of studies on Chinese firms.

Table 1
Sample firm distribution across various industries.

Industry	Quantity	Proportion (%)
Manufacturing industry	62	52.54
Wholesale and retail trade	13	11.02
Warehousing and transportation industry	9	7.63
Realty industry	6	5.08
Business services	6	5.08
Construction business	5	4.24
Metal energy industry	5	4.24
Electric heating industry	4	3.39
Internet industry	2	1.69
Eco-environment industry	2	1.69
Agriculture	2	1.69
Integrated industry	2	1.69

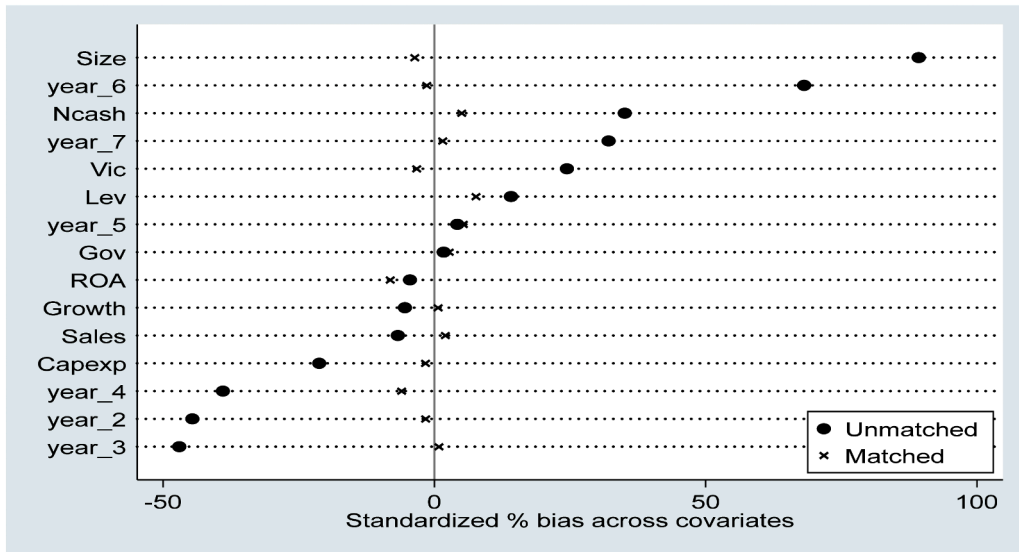


Fig. 1. Bias across covariates.

Fourth, we command “PSgraph” to verify the common support interval. Fig. 2 shows the test results, which are supportive to our matching results.

the year before SCF commitment)/total assets in the year before SCF commitment. Following Fresard (2010), we adopt competitive advantage (*Com*), measured as the difference in growth rate of operation income between the firm and the industry average, to measure the firm's competitiveness in the market.

We also adopt a list of other variables, which include a dummy variable (*SCF*) indicating whether the firm is engaged in SCF, firm size (*Size*), firm internal control quality (*Vic*), firm growth (*Growth*), governance (*Gov*), profitability (*ROA*), leverage (*Lev*), capital expenditure (*Capexp*), non-cash assets (*Ncash*), sales cost (*Sales*), state ownership (*State*), and industry (*Ind*). Table 2 defines the detailed measurement of variables in our primary tests.

To test our hypotheses H1, we develop the following Eq. (1) for empirical test.

$$\Delta Zcash = \alpha_0 + \alpha_1 SCF + \alpha_2 Size + \alpha_3 Vic + \alpha_4 Growth + \alpha_5 Gov + \alpha_6 ROA + \alpha_7 Lev + \alpha_8 Capexp + \alpha_9 Ncash + \alpha_{10} Sales + \alpha_{11} State + \alpha_{12} Ind + \varepsilon \quad (1)$$

As indicated in our H1, we expect the coefficients in Eq. (1) to be positive. In other words, the core firm has more cash holdings after commitment to SCF than that before commitment.

To test our hypothesis H2, we develop the following Eqs. (2) and (3) for empirical test.

$$Com = \beta_0 + \beta_1 Zcash + \beta_2 Size + \beta_3 Vic + \beta_4 Growth + \beta_5 Gov + \beta_6 ROA + \beta_7 Lev + \beta_8 Capexp + \beta_9 Ncash + \beta_{10} Sales + \beta_{11} State + \beta_{12} Ind + \varepsilon \quad (2)$$

$$Com = \beta_0 + \beta_1 \Delta Zcash + \beta_2 Size + \beta_3 Vic + \beta_4 Growth + \beta_5 Gov + \beta_6 ROA + \beta_7 Lev + \beta_8 Capexp + \beta_9 Ncash + \beta_{10} Sales + \beta_{11} State + \beta_{12} Ind + \varepsilon \quad (3)$$

Eqs. (2) and (3) respectively test the relation between *SCF* and *Zcash* and $\Delta Zcash$. As indicated in our hypothesis H2, we expect the coefficients in Eqs. (2) and (3) to be positive. In other words, more cash holdings improve the core firm's competitiveness.

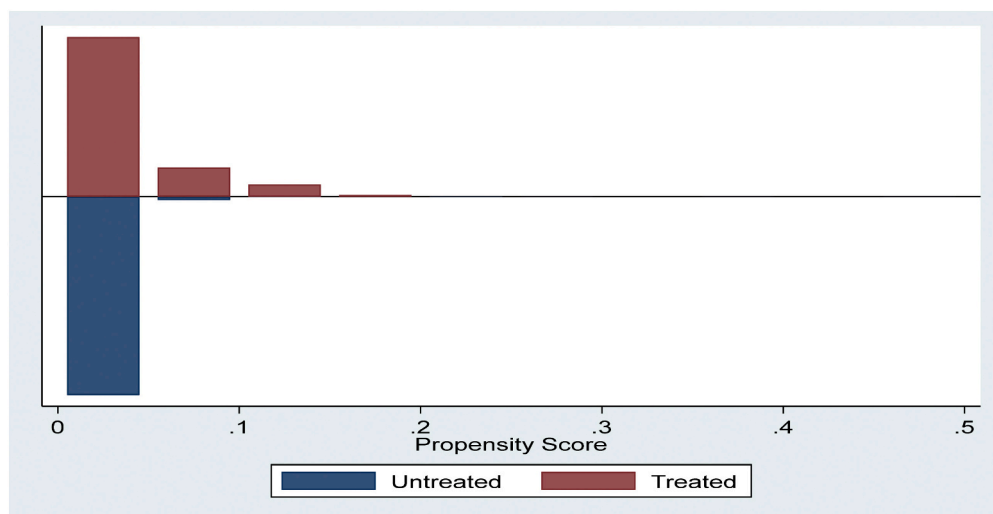


Fig. 2. Common support interval test results.

Following our second command, the statistical software shows PSM results. However, only the PSM results passing tests in our third and fourth commands satisfy our matching requirements. Eventually, we achieve 438 firms without SCF commitment as our comparison samples.

Table 2

Definition of variables in our primary tests.

Variable type	Variable name	Variable definition
Explained variables	Cash holding level change ratio ($\Delta Zcash$)	$\Delta Zcash = (\text{cash holdings 1 year after carrying out Supply Chain Finance} - \text{cash holdings 1 year before carrying out Supply Chain Finance}) / \text{total assets}$
Explanatory variables	Competitive effect(Com)	Increase rate of corporate operating revenue - Industry average
	Whether to carry out the supply chain finance business(SCF)	Dummy variable, carry out takes 1, otherwise takes 0
Control variables	Cash holding ratio(Zcash)	(Cash + short-term investment or transactional financial assets)/total assets
	Firm size(Size)	The total assets scale takes natural logarithm
	Internal control quality(Vic)	This data extracted from DIB internal control and risk management database, Grouping according to the median of the internal control index, take 1 greater than the median, otherwise take 0.
	Enterprise growth(Growth)	Tobin Q value
	Corporate governance level(Gov)	The proportion of independent directors
	Profitability(ROA)	Return on assets = net profit/total assets
	Financial leverage ratio(Lev)	Total liabilities/total assets
	Capital expenditure(Capexp)	(Cash paid for acquisition and construction of fixed assets, intangible assets and other long-term assets - net cash recovered from disposal of fixed assets, intangible assets and other long-term assets)/Total assets
	Non-cash asset(Ncash)	(Total assets-cash assets)/total assets
	Sales expenses(Sales)	Company operating expenses/total assets
	Control right nature(State)	Dummy variable, the value of the state-owned enterprise is 1, and the value of the private enterprise is 0.
	Industry	The manufacturing industry sets the industry dummy variable according to the second level, and the other sets the industry dummy variable according to the first level.

4. Result analysis

4.1. Statistical summary

Table 3 presents the statistical summary of our variables in our primary tests. The mean of *Zcash* is similar between the SCF and comparison groups. The mean of $\Delta Zcash$ of the core firm group is significantly higher than that of the comparison group. Within the core firm group, $\Delta Zcash$ has a minimum of 0, suggesting that no core firm decreases cash holdings in our sample. However, certain firms in the comparison group witness decreases in cash holdings. Interestingly, *Com* has a negative mean value in both groups, suggesting that the sample firm's competitiveness is close to the industrial average. If we consider the time lag on improved firm competitiveness in the market, *Com* values in both groups may not show any clear pattern at this stage. If we ignore the possible time lag of SCF impact, the two groups have similar competitiveness values, further suggesting the comparability of our sample firms. Other variables also suggest that the selection of comparison samples does not have an obvious bias issue.

Table 3
Statistical summary of variables.

Variable	Sample size	Mean	Median	Standard deviation	Min	Max
SCF group						
<i>Com</i>	118	−0.005	−0.025	0.393	−2.747	1.703
$\Delta Zcash$	118	0.143	0.068	0.216	0	0.957
<i>Zcash</i>	118	0.110	0.080	0.090	0.001	0.390
<i>Size</i>	118	22.82	22.84	1.350	20.36	25.57
<i>Vic</i>	118	670.4	677.2	72.42	429.7	830.5
<i>Growth</i>	118	2.010	1.540	1.910	0.190	11.30
<i>Gov</i>	118	0.370	0.330	0.050	0.300	0.500
<i>ROA</i>	118	0.040	0.040	0.060	−0.220	0.190
<i>Lev</i>	118	0.390	0.380	0.190	0.030	0.790
<i>Capexp</i>	118	0.020	0.010	0.020	0	0.130
<i>Ncash</i>	118	0.890	0.910	0.100	0.500	1
<i>Sales</i>	118	0.020	0.010	0.030	0	0.170
<i>State</i>	118	0.400	0	0.490	0	1
Comparison group						
<i>Com</i>	438	−0.109	−0.020	0.572	−3.197	1.703
$\Delta Zcash$	438	0.050	0.010	0.160	−0.310	0.960
<i>Zcash</i>	438	0.111	0.085	0.090	0.001	0.434
<i>Size</i>	438	22.69	22.49	1.300	19.97	25.79
<i>Vic</i>	438	666.0	672.1	77.35	339.9	829.7
<i>Growth</i>	438	2.060	1.400	1.980	0.190	11.30
<i>Gov</i>	438	0.380	0.330	0.060	0.310	0.570
<i>ROA</i>	438	0.050	0.040	0.050	−0.040	0.250
<i>Lev</i>	438	0.400	0.380	0.220	0.010	0.870
<i>Capexp</i>	438	0.010	0.010	0.020	−0.020	0.110
<i>Ncash</i>	438	0.880	0.910	0.110	0.470	1
<i>Sales</i>	438	0.020	0.010	0.040	0	0.270
<i>State</i>	438	0.390	0	0.490	0	1

4.2. Measurement results

Our primary test results from Eqs. (1) and (2) are reported in Table 4. Column (1) results show that *SCF* and $\Delta Zcash$ are significantly and positively related in our full sample, supporting our H1. In addition, *Size* is positively related to $\Delta Zcash$, but *Ncash* and *State* are negatively related to $\Delta Zcash$. These results suggest that larger firms may have more improvement in cash holdings. However, non-cash assets and state ownership tend to be negative towards increases in the firm's cash holdings. Column (2) reports that *Com* is significantly and positively related to *SCF*, supporting our H2. However, *Com* is negatively related to *Size*, *Growth*, *Gov*, *Lev*, and *Ncash* among all sample firms. Columns (3) and (5) respectively suggest that the core firm's *Zcash* and $\Delta Zcash$ are significantly and positively related to *Com*. However, such relations do not exist in the comparison group. In addition, *Size* and *Growth* significantly and negatively contribute to the core firm's *Com*, but without significant relation to the comparison group. *Gov* is significantly and positively related to *Com* in Column (3) but not significant in Column (5). However, *Gov* is significantly and negatively related to *Com* in Columns (4) and (6), suggesting that governance on average improves the core firm's competitiveness, while reducing firm competitiveness in the comparison group. In short, both H1 and H2 are supported by our primary test results.

4.3. Mechanism test

To explore the mechanism underlying our findings in 4.2, we examine how *SCF* works on increasing the core firm's cash holdings and competitiveness. For this purpose, we adopt three operating performance indicators of the core firm in the year before committing to *SCF*. They are accounts receivable turnover (operating income/accounts receivable), accounts payable turnover (operating income/accounts payable), and inventory turnover (operating cost/inventory balance). Using the median of each indicator, we divide our whole sample into high and low groups. In addition, following Farris and Hutchison (2002), we use the cash turnover period (CCC)⁵ as a measurement of capital efficiency in *SCF* and develop the following Eq. (4) for empirical test.

$$CCC = \alpha_0 + \alpha_1 SCF + \alpha_2 Size + \alpha_3 Vic + \alpha_4 Growth + \alpha_5 Gov + \alpha_6 ROA + \alpha_7 Lev + \alpha_8 Capexp + \alpha_9 Ncash + \alpha_{10} Sales + \alpha_{11} State + \varepsilon \quad (4)$$

⁵ CCC has been widely used in management accounting as a key measurement of working capital efficiency. Stewart (1995) defines it as the average number of days required in the conversion of 1 dollar valued raw material to 1 dollar received from customers. CCC has three components, accounts receivable turnover period (payback time of accounts receivable), inventory turnover period (inclusive of in-production and finished products), and accounts payable turnover period. Therefore, CCC = accounts receivable turnover period + inventory turnover period − accounts payable turnover period.

Table 4
Primary test results.

Variables	(1)	(2)	(3)	(4)	(5)	(6)
	Whole sample	Whole sample	SCF group	Comparison group	SCF group	Comparison group
	$\Delta Zcash$	<i>Com</i>	<i>Com</i>	<i>Com</i>	<i>Com</i>	<i>Com</i>
<i>SCF</i>	0.103*** (4.56)	0.103** (2.09)				
<i>Zcash</i>			1.137* (1.68)	−0.360 (−0.83)		
$\Delta Zcash$					0.356* (1.77)	0.229 (1.24)
<i>Size</i>	0.019* (1.88)	−0.036* (−1.66)	−0.094* (−1.72)	−0.016 (−0.65)	−0.110* (−1.86)	−0.020 (−0.80)
<i>Vic</i>	−0.010 (−0.60)	0.012 (0.27)	0.090 (1.07)	−0.014 (−0.25)	0.116 (1.33)	−0.018 (−0.32)
<i>Growth</i>	0.001 (0.15)	−0.065*** (−3.28)	−0.102* (−1.76)	−0.060*** (−2.99)	−0.109* (−1.79)	−0.061*** (−3.04)
<i>Gov</i>	−0.085 (−0.68)	−0.992* (−1.94)	1.398* (1.72)	−1.528*** (−2.67)	1.149 (1.51)	−1.526*** (−2.70)
<i>ROA</i>	0.188 (0.93)	0.087 (0.15)	−0.003 (−0.00)	0.125 (0.16)	0.120 (0.19)	0.022 (0.03)
<i>Lev</i>	0.019 (0.38)	−0.268** (−2.08)	−0.011 (−0.06)	−0.322* (−1.97)	0.158 (0.63)	−0.360** (−2.22)
<i>Capexp</i>	0.157 (0.46)	1.088 (1.30)	2.128 (1.28)	0.632 (0.60)	1.984 (1.19)	0.484 (0.47)
<i>Ncash</i>	−0.268** (−2.34)	−0.395* (−1.82)	1.319* (1.86)	−0.668* (−1.94)	0.700 (1.66)	−0.422 (−1.58)
<i>Sales</i>	−0.242 (−1.02)	0.140 (0.34)	−0.018 (−0.01)	0.297 (0.61)	0.023 (0.02)	0.331 (0.68)
<i>State</i>	−0.029* (−1.71)	−0.071 (−1.35)	−0.087 (−1.23)	−0.058 (−0.86)	−0.051 (−0.69)	−0.051 (−0.75)
<i>Ind</i>	Yes	Yes	Yes	Yes	Yes	Yes
Constant	−0.092 (−0.42)	1.591*** (2.77)	0.694 (0.63)	1.585** (2.12)	1.738 (1.49)	1.409** (2.06)
Observation	556	556	118	438	118	438
Adjusted R ²	0.083	0.123	0.101	0.047	0.102	0.049

Note:***, **and * respectively indicate significance level at the 1%, 5% and 10%. This table reports our primary test results on H1 and H2 through Eqs. (1)–(3). We expect the coefficients in Eqs. (1)–(3) to be positive, i.e. the core firm has more cash holdings after commitment to SCF and more cash holdings improve its competitiveness.

$$\Delta Zcash = \alpha_0 + \alpha_1 SCF + \alpha_2 Size + \alpha_3 Vic + \alpha_4 Growth + \alpha_5 Gov + \alpha_6 ROA + \alpha_7 Lev + \alpha_8 Capexp + \alpha_9 Ncash + \alpha_{10} Sales + \alpha_{11} State + \alpha_{12} Ind + \varepsilon \quad (1)$$

$$Com = \beta_0 + \beta_1 Zcash + \beta_2 Size + \beta_3 Vic + \beta_4 Growth + \beta_5 Gov + \beta_6 ROA + \beta_7 Lev + \beta_8 Capexp + \beta_9 Ncash + \beta_{10} Sales + \beta_{11} State + \beta_{12} Ind + \varepsilon \quad (2)$$

$$Com = \beta_0 + \beta_1 \Delta Zcash + \beta_2 Size + \beta_3 Vic + \beta_4 Growth + \beta_5 Gov + \beta_6 ROA + \beta_7 Lev + \beta_8 Capexp + \beta_9 Ncash + \beta_{10} Sales + \beta_{11} State + \beta_{12} Ind + \varepsilon \quad (3)$$

$\Delta Zcash$ = (cash holdings in the year after SCF commitment – cash holdings in the year before SCF commitment)/total assets in the year before SCF commitment. *Com*, measured as the difference in growth rate of operation income between the firm and the industry average, to measure the firm's competitiveness in the market. Other variables please refer to Table 2.

Table 5 reports our measurement results by Eq. (4). Columns (1) and (2) show significant and positive links between *SCF* and $\Delta Zcash$, however, the coefficient is larger in the sample with lower accounts receivable turnover, suggesting that SCF has a strong impact on shortening the accounts receivable turnover period and enhancing cash holdings at the core firm. Similarly, *SCF* is significantly and positively related to $\Delta Zcash$ in columns (3) to (6), suggesting SCF also affects accounts payable and inventory turnovers and positively contributes to more cash holdings at the core firm. SCF positively contributes to firm cash holdings among both high and low performing firms in our sample, following the above three indicators. However, the coefficient was larger in the lower sample group and differences in the coefficient between groups passed the permutation test. Column (7) shows the significant and negative relation between *SCF* and *CCC*, suggesting that SCF shortens the cash turnover period by around 145 days. Our finding, that SCF significantly improves capital efficiency, is in conformity with that by Lamoureux and Evans (2011) and Ali et al. (2019). Given the outstanding financial constraints among SMEs and private firms in China (Xu et al., 2015; He et al., 2019), such a contribution by SCF may fundamentally change the firm's competitive position in the market.

Table 5
Operating performance and supply chain finance.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	High receivable turnover rate	Low receivable turnover rate	High account payable turnover rate	Low account payable turnover rate	High inventory turnover rate	Low inventory turnover rate	Whole sample
	$\Delta Zcash$	$\Delta Zcash$	$\Delta Zcash$	$\Delta Zcash$	$\Delta Zcash$	$\Delta Zcash$	CCC
SCF	0.079*** (3.25)	0.127*** (3.32)	0.060*** (2.76)	0.150*** (3.94)	0.054*** (2.80)	0.175*** (3.75)	−144.993*** (−3.56)
Size	0.019 (1.47)	0.014 (1.17)	0.012 (0.98)	0.024* (1.74)	0.015 (1.30)	0.022 (1.54)	−10.686 (−0.48)
Vic	0.007 (0.37)	0.004 (0.19)	0.017 (0.94)	−0.010 (−0.42)	0.025 (1.34)	−0.019 (−0.83)	63.594 (1.31)
Growth	−0.002 (−0.19)	−0.001 (−0.12)	−0.002 (−0.26)	0.001 (0.20)	0.004 (0.43)	−0.003 (−0.45)	−14.671 (−1.10)
Gov	0.098 (0.61)	−0.058 (−0.32)	0.035 (0.20)	−0.051 (−0.27)	−0.003 (−0.02)	−0.068 (−0.35)	−138.365 (−0.42)
Roa	0.193 (0.71)	0.010 (0.40)	0.125 (0.52)	0.359 (1.02)	0.099 (0.38)	0.349 (1.16)	28.184 (0.07)
Lev	−0.065 (−1.03)	0.140** (2.23)	−0.009 (−0.17)	0.060 (0.87)	0.004 (0.07)	0.057 (0.83)	480.049*** (3.88)
Capexp	−0.564* (−1.73)	0.656 (1.03)	−0.036 (−0.10)	−0.035 (−0.06)	−0.148 (−0.38)	−0.008 (−0.01)	−2691.079*** (−3.39)
Ncash	−0.421*** (−2.72)	−0.065 (−0.55)	−0.350** (−2.01)	−0.174 (−1.33)	−0.337** (−2.01)	−0.199 (−1.47)	191.559 (1.09)
Sales	−0.461 (−1.55)	0.049 (0.15)	−0.413 (−1.31)	0.026 (0.09)	−0.229 (−0.85)	−0.209 (−0.61)	−734.843* (−1.77)
State	−0.019 (−0.86)	−0.062*** (−2.80)	−0.024 (−1.21)	−0.065*** (−2.66)	−0.019 (−1.05)	−0.068*** (−2.71)	−93.645* (−1.73)
Constant	−0.001 (−0.00)	−0.252 (−0.85)	0.080 (0.31)	−0.340 (−1.18)	−0.007 (−0.03)	−0.244 (−0.81)	303.586 (0.57)
Observations	294	262	293	263	293	263	556
Adjusted R ²	0.120	0.083	0.051	0.102	0.049	0.124	0.068

Note:***, **and * respectively indicate significance level at the1%, 5% and 10%.

This table reports Eq. (4) test results on SCF mechanism.

CCC = $\alpha_0 + \alpha_1 SCF + \alpha_2 Size + \alpha_3 Vic + \alpha_4 Growth + \alpha_5 Gov + \alpha_6 ROA + \alpha_7 Lev + \alpha_8 Capexp + \alpha_9 Ncash + \alpha_{10} Sales + \alpha_{11} State + \varepsilon$ (4)

CCC is the cash turnover period as a measurement of capital efficiency in SCF. Other variables please refer to Table 2.

4.4. Extended discussion

Following our above analysis, we further extend our examination to commercial credit, relationships with banks, political connections, possible over-investment, and firm value of the core firm.

4.4.1. Commercial credit

Commercial credit may be an alternative to bank loans and alleviate financial constraints (Petersen and Rajan, 1997). In a competitive buyer's market, a firm of good reputation can access commercial credit provided by suppliers at a low cost. Such credit not only helps suppliers to achieve better sales but also is meaningful in sustaining the business ties within the supply chain (Fabbri and Menichini, 2010; Giannetti et al., 2011). Mismatches between commercial credit and capital needs may put the supply chain in a risky position. Coordinating capital flows in both upstream and downstream firms, the core firm in SCF may need to effectively manage commercial credit and bank loans to reduce the risks to the supply chain and to reduce the cost of capital, which may occur during the transformation of commercial credit to bank loans.

Given the core firm in SCF may need to manage commercial credit of various values, we further examine the commercial credit impact on the core firm's cash holdings. We use *TradeCredit* as the measurement of commercial credit⁶ and develop the following Eqs. (5) and (6) to examine the role of commercial credit in SCF.

$$SCF = \alpha_0 + \alpha_1 TradeCredit + \alpha_2 Size + \alpha_3 Vic + \alpha_4 Growth + \alpha_5 Gov + \alpha_6 ROA + \alpha_7 Lev + \alpha_8 Capexp + \alpha_9 Ncash + \alpha_{10} Sales + \alpha_{11} State + \alpha_{12} Ind + \varepsilon \quad (5)$$

⁶ Following Ge and Qiu (2007), *TradeCredit* is calculated as the natural logarithm of a firm's aggregate of accounts receivable and accounts payable. Given the data constraints and possible needs of frequent trades with other firms in the supply chain, we do not differentiate between commercial credits from upstream or downstream sources.

$$\Delta Zcash = \alpha_0 + \alpha_1 TradeCredit + \alpha_2 Size + \alpha_3 Vic + \alpha_4 Growth + \alpha_5 Gov + \alpha_6 ROA + \alpha_7 Lev + \alpha_8 Capexp + \alpha_9 Ncash + \alpha_{10} Sales + \alpha_{11} State + \alpha_{12} Ind + \varepsilon \quad (6)$$

Table 6 reports the measurement results of Eqs. (5) and (6). *TradeCredit* is significantly and positively related to *SCF* in the full sample, suggesting that firms with more commercial credit are more likely to commit to *SCF*. In other words, commercial credit is an important factor leading to *SCF*. Columns (2) and (3) suggest that SOEs are more likely to commit to *SCF* than private firms. SOEs are often better supported in various means by the government (Xu and Lin, 2007; He et al., 2019; Li et al., 2020b). Access to more commercial credit does not always induce private firms to commit to *SCF*. This may be largely attributed to the financial constraints of private firms, which require them to be prudential with their commercial credit and better manage their risks in the market. The results in columns (4) and (5) further suggest that, despite significant and positive contributions to firm's cash holdings in both groups, the lower commercial credit group demonstrates stronger impact. In other words, a firm with lower commercial credit tends to hold more cash. This result may be related to the fact that *SCF* only recently started in the country. Given the existence of information asymmetry, banks may tend to lend more to a core firm with more commercial credit. Subsequently, firms with lower commercial credit may have to increase cash holdings to survive in an uncertain market.

4.4.2. Bank relationship

In the largest bank-based economy, the relationship with banks may also contribute to the firm's resources (Levine, 2002; Fu et al., 2015; Xu et al., 2015). A good relationship with banks may also be significant towards *SCF*. To measure the bank-firm relationship, we use a proxy *BC* to indicate whether bank and firm hold each other's shares or whether the firm executive has a banking background.⁷ We examine the role of bank relationships with the following Eq. (7).

$$\Delta Zcash = \alpha_0 + \alpha_1 SCF + \alpha_2 BC + \alpha_3 BC \times SCF + \alpha_4 Growth + \alpha_5 Gov + \alpha_6 ROA + \alpha_7 Lev + \alpha_8 Capexp + \alpha_9 Ncash + \alpha_{10} Sales + \alpha_{11} State + \alpha_{12} Ind + \varepsilon \quad (7)$$

Table 7 reports the measurement results of Eq. (7). *BC* and *SCF* are significantly and positively related in the full sample, suggesting that bank relationship is an important external factor in the firm to committing to *SCF*. However, there is no significant link identified between *BC* or *BC* $\times$ *SCF* and $\Delta Zcash$. In other words, good bank relationships do not necessarily contribute to improved cash holdings among sample firms. The examination of firm ownership shows that *BC* and $\Delta Zcash$ are significantly and positively related in the SOE group but not in the private firm group. However, *BC* $\times$ *SCF* and $\Delta Zcash$ are significantly and positively related in the private firm group. These results suggest that state ownership still decides a firm's access to bank loans, which is significant towards the firm's cash holdings among SOEs. Good bank relationships and *SCF* may not always result in improved cash holdings within SOEs. When both good bank relationships and *SCF* exist, private firms may show significantly improved cash holdings. In short, good bank relationships may contribute to more cash holdings in the core firm.

4.4.3. Political connections

Given that political connections in China often play an important role in such firm policies as cash holdings (Faccio, 2006; Liu and Tian, 2012; Xu et al., 2016; He et al., 2019), we further examine the effect of political connections on the core firm's cash holdings. Considering that the president and CEO are decisive members of the senior management in firm operation, we follow Fan et al. (2007) in defining firm political connections and develop the following Eq. (8) for such a measurement.

$$\Delta Zcash = \alpha_0 + \alpha_1 SCF + \alpha_2 POL + \alpha_3 POL \times SCF + \alpha_4 Growth + \alpha_5 Gov + \alpha_6 ROA + \alpha_7 Lev + \alpha_8 Capexp + \alpha_9 Ncash + \alpha_{10} Sales + \alpha_{11} State + \alpha_{12} Ind + \varepsilon \quad (8)$$

Where *POL* is the political connection by the firm's president and CEO, indicating whether he or she has such experience as a government official, an army officer, standing committee member or member of the People's Congress or CPPCC. If yes, *POL* equals to 1, otherwise 0.

Table 8 reports our test results on political connections. Among all sample firms, *POL* is significantly and positively related to $\Delta Zcash$, confirming that political connections are significant towards the firm's cash holdings. Furthermore, *POL* $\times$ *SCF* is significantly and positively related to $\Delta Zcash$, suggesting that political connections have positive impacts on the core firm's cash holdings. Political connections may signal better reputation and support from the government towards creditors and towards the *SCF*, and subsequently allow the core firm to improve its cash holdings. However, such an impact only exists among the privately owned core firms, not the state-owned ones. In other words, since SOEs naturally enjoy political connections, participating in *SCF* would not noticeably improve their cash holdings. In short, private ownership with political connections may further improve the core firm's cash holdings.

4.4.4. Firm investment

It can be interesting to examine the core firm's over-investment issue when *SCF* contributes to more cash holdings. Following the literature discussion in Part 2, we further examine the consequence of increased cash holdings at the core firm on its investment. Following Richardson (2006), we develop the following models to examine firm's possible over-investment.

$$Inv_{i,t} = \alpha_0 + \alpha_1 Inv_{i,t-1} + \alpha_2 Tobin'sQ + \alpha_3 CFO_{i,t-1} + \alpha_4 ROA_{i,t-1} + \alpha_5 Size_{i,t-1} + \varepsilon_{i,t-1} \quad (9)$$

Where *Inv* is the firm's investment, measured as (the net changes in fixed assets, building-in-progress, intangible assets, and long-term investments)/total assets. *CFO* is the net cash flow from operations. Then the residual in Eq. (9) is used as the over-investment

⁷ *BC* is a dummy variable. It equals to 1 if the bank and firm holds each other's shares or a firm executive has a banking background, otherwise, 0.

Table 6
Commercial credit test results.

Variable	(1)	(2)	(3)	(4)	(5)
	Whole sample	SOEs	Private firms	High commercial credit scale	Low commercial credit scale
	SCF	SCF	SCF	$\Delta Zcash$	$\Delta Zcash$
<i>Trade-Credit</i>	0.112*** (3.23)	0.312*** (2.80)	0.071 (1.64)		
<i>SCF</i>				0.052** (2.51)	0.136*** (3.82)
<i>Size</i>	0.210 (1.49)	0.253 (1.02)	0.126 (0.54)	0.004 (0.29)	0.034*** (2.65)
<i>Vic</i>	-0.106 (-0.42)	-0.365 (-0.77)	-0.273 (-0.81)	0.012 (0.73)	-0.018 (-0.72)
<i>Growth</i>	0.094 (1.19)	0.112 (0.36)	0.094 (0.99)	0.011 (1.11)	-0.005 (-0.89)
<i>Gov</i>	-1.191 (-0.53)	-2.170 (-0.47)	0.308 (0.10)	-0.092 (-0.72)	0.166 (0.71)
<i>ROA</i>	-2.486 (-0.89)	9.132 (1.23)	-2.753 (-0.72)	0.207 (0.80)	0.281 (1.01)
<i>Lev</i>	-1.311* (-1.86)	-2.371 (-1.58)	-0.948 (-0.94)	0.040 (0.76)	0.032 (0.47)
<i>Capexp</i>	-2.509 (-0.44)	-22.913* (-1.68)	1.966 (0.21)	-0.297 (-0.88)	0.434 (0.77)
<i>Ncash</i>	2.078 (1.55)	4.187 (1.17)	2.211 (1.28)	-0.208* (-1.73)	-0.325** (-2.17)
<i>Sales</i>	4.207 (1.13)	16.710* (1.83)	-1.091 (-0.25)	-0.137 (-0.63)	-0.699* (-1.67)
<i>State</i>	-0.295 (-1.03)			-0.026 (-1.56)	-0.057** (-2.22)
<i>Ind</i>	Yes	Yes	Yes	No	No
Constant	-8.919*** (-2.59)	-15.848** (-2.57)	-7.538 (-1.28)	0.160 (0.73)	-0.488 (-1.56)
Observations	556	217	339	278	278
Pseudo R ²	0.127	0.265	0.151		
Adjusted R ²				0.035	0.131

Note:***, **and * respectively indicate significance level at the1%, 5% and 10%.

This table reports our test results through Eqs. (5)–(6).

$SCF = \alpha_0 + \alpha_1 TradeCredit + \alpha_2 Size + \alpha_3 Vic + \alpha_4 Growth + \alpha_5 Gov + \alpha_6 ROA + \alpha_7 Lev + \alpha_8 Capexp + \alpha_9 Ncash + \alpha_{10} Sales + \alpha_{11} State + \alpha_{12} Ind + \varepsilon$ (5)

$\Delta Zcash = \alpha_0 + \alpha_1 TradeCredit + \alpha_2 Size + \alpha_3 Vic + \alpha_4 Growth + \alpha_5 Gov + \alpha_6 ROA + \alpha_7 Lev + \alpha_8 Capexp + \alpha_9 Ncash + \alpha_{10} Sales + \alpha_{11} State + \alpha_{12} Ind + \varepsilon$ (6)

TradeCredit as the measurement of commercial credit. Other variables please refer to Table 2.

index (*OverInv*)⁸ for examination through following Eqs. (10)–(12).

$$OverInv_{i,t} = \beta_0 + \beta_1 Zcash_{i,t} + \beta_2 Controls_{i,t-1} + \varepsilon_{i,t} \quad (10)$$

$$OverInv_{i,t} = \beta_0 + \beta_1 \Delta Zcash_{i,t} + \beta_2 Controls_{i,t-1} + \varepsilon_{i,t} \quad (11)$$

$$OverInv_{i,t} = \beta_0 + \beta_1 FCF_{i,t} + \beta_2 Controls_{i,t-1} + \varepsilon_{i,t} \quad (12)$$

In the above Eqs. (10)–(12), *OverInv* is the over-investment by the core firm. A smaller value in *OverInv* corresponds to better investment efficiency and less likelihood of over-investment. *Controls* refers to the control variables of *Size*, *Vic*, *Growth*, etc. that we use in previous examinations. *FCF* is the firm's free cash flow, measured as (net profit + depreciation – common share dividends)/total assets.

Table 9 reports our measurement results on the core firm's investment. *Zcash* and $\Delta Zcash$ are significantly and negatively related to *OverInv*, suggesting that increased cash holdings at the core firm may not lead to over-investment. On the contrary, the more cash holdings the core firm has, the higher investment efficiency may be achieved. One possible explanation can be that SCF effectively alleviates the core firm's financial constraints and subsequently allows improved investment efficiency. In other words, SCF may

⁸ Considering the small sample of core firms in SCF, we adopt the absolute value of the residual in Eq. (8) as *OverInv*, where positive coefficients indicate low investment efficiency and possible over-investment, while negative coefficients indicate investment efficiency proportional to a firm's cash holding and no over-investment.

Table 7
Bank relationship test results.

Variables	(1)	(2)	(3)	(4)
	Whole sample	Whole sample	SOEs	Private firms
	SCF	$\Delta Zcash$	$\Delta Zcash$	$\Delta Zcash$
SCF		0.074** (2.34)	0.179*** (2.94)	0.034 (0.88)
BC	0.479* (1.67)	0.012 (0.65)	0.033* (1.78)	−0.006 (−0.20)
BC × SCF		0.035 (0.87)	−0.086 (−1.25)	0.094* (1.76)
Size	0.144 (1.26)	0.018* (1.83)	−0.009 (−1.14)	0.031 (1.60)
Vic	0.054 (0.25)	−0.008 (−0.51)	−0.009 (−0.52)	−0.012 (−0.43)
Growth	0.015 (0.17)	0.001 (0.10)	−0.006 (−0.54)	0.003 (0.37)
Gov	−1.341 (−0.75)	−0.092 (−0.72)	−0.224 (−1.35)	−0.018 (−0.09)
ROA	−3.115 (−1.11)	0.180 (0.88)	0.329 (1.19)	0.093 (0.32)
Lev	−0.757 (−1.26)	0.021 (0.43)	−0.022 (−0.30)	0.051 (0.61)
Capexp	2.589 (0.53)	0.166 (0.48)	0.093 (0.15)	−0.379 (−1.16)
Ncash	1.122 (0.96)	−0.268** (−2.34)	−0.176 (−1.32)	−0.323** (−2.02)
Sales	1.156 (0.46)	−0.250 (−1.05)	0.089 (0.30)	0.206 (0.39)
State	−0.092 (−0.38)	−0.030* (−1.76)		
Ind	No	Yes	Yes	Yes
Constant	−5.109* (−1.76)	−0.088 (−0.40)	0.390* (1.79)	−0.316 (−0.73)
Observations	556	556	217	339
Adjusted R ²	0.014	0.082	0.251	0.054

Note:***, **and * respectively indicate significance level at the1%, 5% and 10%. This table reports the measurements results of Eq. (7).

$\Delta Zcash = \alpha_0 + \alpha_1 SCF + \alpha_2 BC + \alpha_3 BC \times SCF + \alpha_4 Growth + \alpha_5 Gov + \alpha_6 ROA + \alpha_7 Lev + \alpha_8 Capexp + \alpha_9 Ncash + \alpha_{10} Sales + \alpha_{11} State + \alpha_{12} Ind + e$

(7)
BC to indicate whether bank and firm holds each other's shares or whether the firm executive has a banking background. Other variables please refer to Table 2.

liberate the core firm to make better investments and achieve enhanced economic benefits. This may further support our primary test findings that improved cash holdings by the core firm is positive towards its competitiveness in the market.

4.4.5. Firm value

It can also be meaningful to explore the changes in a core firm's value when SCF improves its cash holdings. On the one hand, improved cash holdings may allow the firm to take advantage of good investment opportunities, respond quickly to demands from the market, gain better growth opportunities, and subsequently enhance the firm value. On the other, with improved cash holdings, the core firm may better cope with risks in the SCF ecosystem, which is needed for improved firm values. Considering that economic value added and Tobin's Q⁹ may both be used to measure firm value, we develop Eqs. (13) and (14) in our additional tests.

$$\text{Tobin's } Q = \alpha_0 + \alpha_1 SCF + \alpha_2 Zcash + \alpha_3 Zcash \times SCF + \alpha_4 Gov + \alpha_5 ROA + \alpha_6 Lev + \alpha_7 Capexp + \alpha_8 Ncash + \alpha_9 Sales + \alpha_{10} State + \alpha_{11} Ind + \varepsilon. \quad (13)$$

$$EVA = \alpha_0 + \alpha_1 SCF + \alpha_2 Zcash + \alpha_3 Zcash \times SCF + \alpha_4 Gov + \alpha_5 ROA + \alpha_6 Lev + \alpha_7 Capexp + \alpha_8 Ncash + \alpha_9 Sales + \alpha_{10} State + \alpha_{11} Ind + \varepsilon \quad (14)$$

Table 10 reports our test results on firm values. SCF × Z-cash is significantly and positively related to all firm value measurements, suggesting that improved cash holdings at the core firm also enhance the firm values.

⁹ Gomm (2010) suggests that EVA may also be appropriate to measure firm value in SCF context. Accordingly, we respectively use EVA = (the core firm's economic value added)/(the core firm's average net assets) and Tobin's Q in our models Eqs. (13) and (14). In addition, we use $\Delta \text{Tobin's } Q$ and ΔEVA with a time lag to illustrate the changes in firm value.

Table 8
Political connections test results.

Variables	(1)	(2)	(3)	(4)
	Whole sample	Whole sample	SOEs	Private firms
	$\Delta Zcash$	$\Delta Zcash$	$\Delta Zcash$	$\Delta Zcash$
<i>SCF</i>		0.086*** (4.68)	0.106*** (4.85)	0.061** (2.01)
<i>POL</i>	0.129*** (7.95)	0.120*** (7.57)	0.056*** (2.65)	0.151*** (6.63)
<i>POL</i> × <i>SCF</i>		0.067* (1.85)	−0.002 (−0.04)	0.094* (1.74)
<i>Size</i>	0.024*** (2.92)	0.021*** (2.60)	−0.006 (−0.75)	0.027** (2.03)
<i>Vic</i>	−0.013 (−0.86)	−0.011 (−0.74)	−0.007 (−0.41)	−0.019 (−0.80)
<i>Growth</i>	0.004 (0.69)	0.002 (0.44)	−0.009 (−0.91)	0.005 (0.68)
<i>Gov</i>	−0.114 (−0.86)	−0.095 (−0.74)	−0.206 (−1.33)	−0.079 (−0.40)
<i>ROA</i>	0.130 (0.76)	0.187 (1.11)	0.395* (1.83)	0.077 (0.31)
<i>Lev</i>	0.009 (0.21)	0.019 (0.46)	−0.026 (−0.59)	0.042 (0.62)
<i>Capexp</i>	0.026 (0.07)	−0.002 (−0.00)	0.143 (0.32)	0.033 (0.07)
<i>Ncash</i>	−0.221*** (−3.04)	−0.247*** (−3.48)	−0.128 (−1.44)	−0.324*** (−3.16)
<i>Sales</i>	−0.181 (−0.88)	−0.235 (−1.17)	0.142 (0.44)	−0.429 (−1.51)
<i>State</i>	−0.008 (−0.43)	−0.003 (−0.20)		
<i>Ind</i>	Yes	Yes	Yes	Yes
Constant	−0.256 (−1.27)	−0.133 (−0.68)	0.296 (1.30)	−0.194 (−0.58)
Observations	556	556	217	339
Adjusted R2	0.138	0.180	0.267	0.184

Note:***, **and * respectively indicate significance level at the1%, 5% and 10%. This table reports our test results through Eq. (8).

$$\Delta Zcash = \alpha_0 + \alpha_1 SCF + \alpha_2 POL + \alpha_3 POL \times SCF + \alpha_4 Growth + \alpha_5 Gov + \alpha_6 ROA + \alpha_7 Lev + \alpha_8 Capexp + \alpha_9 Ncash + \alpha_{10} Sales + \alpha_{11} State + \alpha_{12} Ind + \varepsilon \quad (8)$$

POL is the political connection by the firm's president and CEO, indicating whether he or she has such experience as a government official, an army officer, standing committee or member of the People's Congress or CPPCC. If yes, *POL* equals to 1, otherwise 0. Other variables please refer to Table 2.

4.5. Robustness

To ensure reliability of our findings, we redo PSM at 1:3 ratio. In the new sample, we use the core firm's cash holdings 1 year after SCF commitment minus its cash holdings in the year of SCF commitment to calculate $\Delta Zcash1$ to replace $\Delta Zcash$. Then we repeat our primary tests and find that SCF is still significantly and positively related to the core firm's cash holdings. Similarly, increased cash holdings are significantly and positively related to the firm's competitiveness. Then we repeat our examination on commercial credit, bank relationship, and political connections by the core firm and get similar results. Then we change the measurement of *Inv* to [(changes in original value of fixed assets + changes in original value of long-term investments)/total assets − depreciation and amortization] and redo the test on the core firm's investment. The results are supportive to our finding. Tests on firm value also show similar results. These results are reported in Table 11. In short, our findings are robust to the new measurements. Given the features of PSM tests, the endogeneity issue is further mitigated in this process.

Table 9
Over-investment test results.

Variables	(1)	(2)	(3)
	<i>Over-inv</i>	<i>Over-inv</i>	<i>Over-inv</i>
<i>Zcash</i>	−0.030** (−2.22)		
$\Delta Zcash$		−0.012*** (−2.69)	
<i>FCF</i>			−0.013 (−0.74)
<i>Size</i>	−0.001 (−0.17)	0.002 (0.33)	0.001 (0.07)
<i>Vic</i>	−0.004* (−1.76)	−0.005** (−2.09)	−0.004* (−1.88)
<i>Growth</i>	0.001 (0.13)	0.002 (0.35)	−0.005 (−0.65)
<i>Gov</i>	0.053 (1.61)	0.058* (1.78)	0.020 (1.01)
<i>ROA</i>	−0.013 (−0.64)	−0.012 (−0.58)	−0.022 (−1.44)
<i>Lev</i>	−0.007 (−1.18)	−0.011* (−1.70)	−0.008 (−1.47)
<i>Capexp</i>	0.063 (0.83)	0.064 (0.80)	0.066 (1.45)
<i>Ncash</i>	−0.043** (−2.44)	−0.024* (−1.70)	−0.013 (−1.15)
<i>Sales</i>	−0.023 (−0.66)	−0.011 (−0.30)	−0.027 (−0.76)
<i>State</i>	0.002 (0.61)	0.001 (0.08)	−0.001 (−0.27)
<i>Ind</i>	No	No	No
Constant	0.040 (1.46)	0.009 (0.36)	0.022 (0.86)
Observations	118	118	118
Adjusted R ²	0.056	0.068	0.228

Note:***, **and * respectively indicate significance level at the1%, 5% and 10%.

This reports our Eqs. (10)–(12) test results.

$$Inv_{i,t} = \alpha_0 + \alpha_1 Inv_{i,t-1} + \alpha_2 Tobin'sQ + \alpha_3 CFO_{i,t-1} + \alpha_4 ROA_{i,t-1} + \alpha_5 Size_{i,t-1} + \varepsilon_{i,t-1} \quad (9)$$

$$OverInv_{i,t} = \beta_0 + \beta_1 Zcash_{i,t} + \beta_2 Controls_{i,t-1} + \varepsilon_{i,t} \quad (10)$$

$$OverInv_{i,t} = \beta_0 + \beta_1 \Delta Zcash_{i,t} + \beta_2 Controls_{i,t-1} + \varepsilon_{i,t} \quad (11)$$

$$OverInv_{i,t} = \beta_0 + \beta_1 FCF_{i,t} + \beta_2 Controls_{i,t-1} + \varepsilon_{i,t} \quad (12)$$

Inv is the firm's investment. *CFO* is the net cash flow from operations. The residual in Eq. (9) is used to determine over-investment index (*OverInv*) in Eq. (10)–(12). Other variables please refer to Table 2.

5. Conclusion

Through the largest sample possible for this empirical study, we find that SCF positively contributes to a core firm's cash holdings and improves its competitiveness. The mechanism tests suggest that SCF enhances both operating performance and cash turnover of the firm. In addition, such an impact is stronger when the firm is of lower efficiency. Our extended examinations indicate that the firm with more commercial credit or of state ownership will more likely become committed to SCF. Good relationships with banks or political connections may further improve the private firm's cash holdings. Furthermore, increased cash holdings in the core firm may lead to enhanced investment efficiency, rather than over-investment, and enhanced firm values. Therefore, SCFs contribution to the core firm may go beyond the improved cash holdings and structurally change firm behaviour in the future.

As a meaningful financial innovation in recent years, SCF may have become a strategic choice for firms in the supply chain. However, the reform in the country's financial market may need to accommodate such changes in the foreseeable future. In our examination of SCF impact on firm cash holdings, we could still identify bias towards private ownership. There may need to be more

Table 10
Firm value test results.

Variables	(1)	(2)	(3)	(4)
	Whole sample	Whole sample	Whole sample	Whole sample
	<i>Tobin Q</i>	Δ <i>Tobin Q</i>	<i>EVA</i>	Δ <i>EVA</i>
<i>SCF</i>	0.519*** (5.80)	0.270*** (3.37)	0.043*** (4.08)	0.034*** (3.06)
<i>Z-cash</i>	−0.153 (−0.29)	1.214*** (2.61)	0.190*** (3.13)	0.169*** (2.64)
<i>SCF</i> × <i>Z-cash</i>	2.150** (2.29)	2.656*** (3.16)	0.228** (2.08)	0.228** (1.98)
<i>Size</i>	0.164*** (4.78)	−0.190*** (−6.20)	0.004 (1.09)	0.001 (0.18)
<i>Vic</i>	−0.065 (−0.86)	0.022 (0.32)	0.025*** (2.88)	−0.002 (−0.17)
<i>Gov</i>	1.121* (1.76)	0.231 (0.40)	0.034 (0.45)	0.048 (0.61)
<i>ROA</i>	−0.292 (−0.35)	1.999*** (2.67)	0.304*** (3.11)	−0.521*** (−5.06)
<i>Lev</i>	0.104 (0.52)	−0.423** (−2.36)	−0.031 (−1.33)	−0.030 (−1.23)
<i>Capexp</i>	0.367 (0.22)	2.789* (1.87)	0.305 (1.57)	−0.033 (−0.16)
<i>Ncash</i>	−0.118 (−0.27)	−0.205 (−0.52)	0.059 (1.16)	0.059 (1.11)
<i>Sales</i>	−0.455 (−0.46)	3.492*** (3.95)	0.224* (1.95)	0.245** (2.02)
<i>State</i>	−0.130 (−1.54)	−0.009 (−0.12)	−0.010 (−1.07)	0.001 (0.14)
<i>Ind</i>	Yes	Yes	Yes	Yes
Constant	−4.149*** (−4.77)	6.117*** (7.86)	−0.176* (−1.74)	0.004 (0.04)
Observations	556	556	556	556
Adjusted R2	0.185	0.334	0.157	0.116

Note:***, **and * respectively indicate significance level at the 1%, 5% and 10%.

This table reports our Eqs. (13)–(14) test results.

$Tobin's Q = \alpha_0 + \alpha_1 SCF + \alpha_2 Zcash + \alpha_3 Zcash \times SCF + \alpha_4 Gov + \alpha_5 ROA + \alpha_6 Lev + \alpha_7 Capexp + \alpha_8 Ncash + \alpha_9 Sales + \alpha_{10} State + \alpha_{11} Ind + \varepsilon$ (13)

$EVA = \alpha_0 + \alpha_1 SCF + \alpha_2 Zcash + \alpha_3 Zcash \times SCF + \alpha_4 Gov + \alpha_5 ROA + \alpha_6 Lev + \alpha_7 Capexp + \alpha_8 Ncash + \alpha_9 Sales + \alpha_{10} State + \alpha_{11} Ind + \varepsilon$ (14)

Tobin's Q and EVA are used as indicators of firm values. Other variables please refer to Table 2.

supportive policies in place to allow SCF to play its positive roles in sustaining the economic growth of the largest bank-based economy. Given that a growing number of listed firms commit to SCF, future studies may provide further evidence on SCF's contribution to the real economy.

Author statement

The authors do not have permission to share data.

Declaration of competing interest

None.

Table 11
Robustness test results.

Panel A										
Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Whole sample	Whole sample	Whole sample	SCF group	Comparison group	SCF group	Comparison group	Whole sample	SOEs	Private firms
	$\Delta Zcash1$	$\Delta Zcash$	SCF	Com	Com	Com	Com	$\Delta Z-cash$	$\Delta Z-cash$	$\Delta Z-cash$
SCF	0.050*** (5.67)	0.111*** (4.07)						0.094*** (4.77)	0.099*** (4.04)	0.067** (2.06)
Trade-Credit			0.114*** (3.16)							
Zcash				1.589** (2.60)	−0.491 (−0.87)					
$\Delta Zcash$						0.264* (1.87)	0.068 (0.53)			
POL								0.127*** (7.02)	0.056** (2.24)	0.161*** (6.34)
POL _{SCF}								0.064* (1.65)	−0.060 (−1.04)	0.096* (1.67)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	0.092 (0.86)	0.204 (0.54)	−3.587 (−1.16)	0.125 (0.14)	−0.256 (−0.26)	1.283 (1.36)	−0.682 (−0.86)	−0.126 (−0.54)	0.268 (1.02)	0.096 (0.24)
Observations	472	472	472	118	354	118	354	452	171	281
Adjusted R2	0.097	0.091		0.131	0.078	0.121	0.069	0.193	0.308	0.199
Pseudo R2			0.139							

Panel B						
Variables	(1)	(2)	(3)	(4)	(5)	(6)
	Experi-metal group	Experi-metal group	Whole sample	Whole sample	Whole sample	Whole sample
	Over-inv	Over-inv	Tobin Q	Δ Tobin Q	EVA	Δ EVA
FCF	−0.012 (−0.77)					
SCF			0.253*** (3.07)	0.403*** (4.25)	0.045*** (3.31)	0.035*** (2.88)
Z-cash		−0.026** (−2.39)	1.091** (2.10)	−0.143 (−0.24)	0.243*** (2.80)	0.181** (2.38)
SCF × Z-cash			2.977*** (3.41)	5.644*** (5.62)	0.287** (1.97)	0.242* (1.90)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Ind	Yes	Yes	Yes	Yes	Yes	Yes
Constant	0.038* (1.69)	0.023 (1.11)	5.239*** (5.82)	−3.409*** (−3.29)	−0.414*** (−2.75)	−0.029 (−0.22)
Observations	118	118	452	452	452	452
Adjusted R2	0.079	0.208	0.349	0.201	0.129	0.0803

Note:***, **and * respectively indicate significance level at the 1%, 5% and 10%. This table reports our robustness test results. We redo PSM at 1:3 ratio. In the new sample, we use the core firm's cash holdings 1 year after SCF commitment minus its cash holdings in the year of SCF commitment to calculate $\Delta Zcash1$ to replace $\Delta Zcash$. Then we repeat our primary tests and find that SCF is still significantly and positively related to the core firm's cash holdings. Similarly, increased cash holdings are significantly and positively related to the firm's competitiveness. Then we repeat our additional tests and get similar results. Even when we change the measurement of *Inv* to [(changes in original value of fixed assets + changes in original value of long-term investments)/total assets – depreciation and amortization], the results are supportive to our findings.

Acknowledgements

This research was supported by the National Social Science Fund Major Project (19AGL010), National Social Science Fund Youth Project (18CJY027), Shandong University Humanities and Social Youth Team Project (IFYT17041), Shandong Provincial Natural Science Foundation, China (ZR2019MG004).

Appendix A. Propensity score matching (PSM) method

To mitigate the issues of selection bias and functional form misspecification (FFM) in our measurements of supply chain finance and firm cash holdings, we adopt PSM method in the construction of comparison group samples. Setting *SCF* values of 1 and 0, corresponding to the firm's commitment to SCF or not, we apply matching variables of *Size*, *Vic*, *Gov*, *ROA*, *Lev*, *Capexp*, *Ncash*, *Sales*, and *year*. At the ratio of 1:4, we use the nearest neighbor matching, to select firms that have the smallest difference in propensity score as that of the core firms in the SCF group. The following are the detailed procedures.

First, we command “set seed 10101”, “Gen TMP = Runiform()” and “sort TMP” are adopted to randomly sort all observations, in statistical software Stata15.

Second, we command “PSmatch2 Treat Size Vic Growth Gov ROA Lev Capexp Ncash Sales year_2 year_3 year_4 year_5 year_6 year_7,out(Zcash) logit neighbor(4) Common caliper(0.05) ties” and achieve the following Tables A-C. Table A illustrates our matching results. Then we judge the matching effect by ATT and *t*-test values. ATT = -0.004 and *t* = -0.4 is less than the critical value of 1.69 (see Table B), suggesting that *t*-test value conforms to matching requirement by PSM. Similarly, SCF and comparison group firms are in the range of common values (See Table C).

Table A
Matching results.

Logistic regression Log likelihood = -566.45423			Number of obs = 12,991 LR chi2(15) = 211.53 Prob > chi2 = 0.0000 Pseudo R2 = 0.1573			
SCF	Coef.	Std.Err.	z	P > z	[95%Conf.	Interval]
Size	0.578	0.075	7.700	0.000	0.431	0.725
Vic	0.002	0.001	2.000	0.046	0.000	0.004
Growth	0.003	0.014	0.230	0.815	-0.024	0.030
Gov	-1.884	1.621	-1.160	0.245	-5.062	1.293
ROA	-0.808	0.681	-1.190	0.235	-2.142	0.526
Lev	-0.269	0.457	-0.590	0.556	-1.165	0.627
Capexp	-2.094	1.114	-1.880	0.060	-4.277	0.088
Ncash	0.808	0.920	0.880	0.380	-0.996	2.612
Sales	2.490	2.241	1.110	0.267	-1.903	6.882
year_2	12.052	582.861	0.020	0.984	-1130.335	1154.439
year_3	12.009	582.861	0.020	0.984	-1130.378	1154.396
year_4	12.626	582.861	0.020	0.983	-1129.760	1155.013
year_5	14.112	582.861	0.020	0.981	-1128.274	1156.498
year_6	15.015	582.861	0.030	0.979	-1127.371	1157.401
year_7	14.484	582.861	0.020	0.980	-1127.902	1156.870
_cons	-32.758	582.864	-0.060	0.955	-1175.150	1109.635

Table B
ATT and T-test values.

Variable	Sample	Treated	Controls	Difference	S.E.	T-stat
Zcash	Unmatched	0.109	0.139	-0.030	0.012	-2.500
	ATT	0.109	0.113	-0.004	0.011	-0.400

Note: S.E. does not include the estimated propensity scores.

Table C
Common value range.

psmatch2:		
psmatch2: Treatment assignment	Common support <i>On support</i>	Total
Untreated	12,873	12,873
Treated	118	118
Total	12,991	12,991

Third, we command “pctest, Both Graph” to test the balance of the matched variables. Table D reports that the bias of all matched variables is less than 10%. Therefore, the matched variables are acceptable. P-values of t-test for all variables after matching are greater than 0.1, confirming that there is no systematic difference in matching variables between the SCF and the comparison groups. Table E and Fig. 1 show the variation of standardized bias after variable matching and the matched data is balanced.

Table D
Balance of matched variables test results.

Variable	Unmatched Matched	Mean		%bias	%reduct bias	t	P > t	V(T)/ V(C)
		Treated	Control					
Size	U	22.809	21.675	89.3		10.60	0.000	1.42
	M	22.809	22.856	−3.7	95.9	−0.26	0.797	0.96
Vic	U	669.99	642.94	24.4		2.16	0.030	0.34*
	M	669.99	673.64	−3.3	86.5	−0.33	0.740	0.77
Growth	U	2.1997	2.7225	−5.5		−0.43	0.666	0.07*
	M	2.1997	2.1382	0.6	88.2	0.16	0.869	2.17*
Gov	U	0.37293	0.37201	1.7		0.18	0.859	0.88
	M	0.37293	0.37147	2.7	−59.0	0.20	0.845	0.72
ROA	U	0.03826	0.0432	−4.5		−0.42	0.676	0.46*
	M	0.03826	0.04721	−8.2	−81.2	−0.90	0.367	1.83*
Lev	U	0.38966	0.3519	14.1		1.28	0.202	0.39*
	M	0.38966	0.36926	7.6	46.0	0.75	0.456	0.84
Capexp	U	0.01498	0.02586	−21.2		−1.76	0.078	0.17*
	M	0.01498	0.01585	−1.7	92.1	−0.23	0.818	0.87
Ncash	U	0.88643	0.83835	35.1		3.35	0.001	0.56*
	M	0.88643	0.87957	5.0	85.7	0.47	0.638	1.17
Sales	U	0.02015	0.02262	−6.8		−0.68	0.500	0.71
	M	0.02015	0.01942	2.0	70.5	0.14	0.888	0.55*
year_2	U	0.01695	0.13105	−44.6		−3.67	0.000	.
	M	0.01695	0.02119	−1.7	96.3	−0.24	0.813	.
year_3	U	0.01695	0.14022	−47.0		−3.85	0.000	.
	M	0.01695	0.01483	0.8	98.3	0.13	0.897	.
year_4	U	0.0339	0.14255	−39.0		−3.37	0.001	.
	M	0.0339	0.05085	−6.1	84.4	−0.64	0.520	.
year_5	U	0.16102	0.14581	4.2		0.47	0.641	.
	M	0.16102	0.14195	5.3	−25.4	0.41	0.684	.
year_6	U	0.45763	0.15901	68.1		8.80	0.000	.
	M	0.45763	0.46398	−1.5	97.9	−0.10	0.922	.
year_7	U	0.31356	0.17688	32.1		3.86	0.000	.
	M	0.31356	0.3072	1.5	95.3	0.11	0.916	.

Note: * if variance ratio outside [0.69; 1.44] for U and [0.69; 1.44] for M.

Table E
Comparison of the matched and unmatched samples.

Sample	Ps R2	LR chi2	P > chi2	MeanBias	MedBias	B	R	%Var
Unmatched	0.158	212.24	0.000	29.2	24.4	112.4*	0.11*	67
Matched	0.008	2.50	1.000	3.4	2.7	20.5	0.90	33

Note: if B > 25%, R outside [0.5; 2].

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